Neuroinformatics Project
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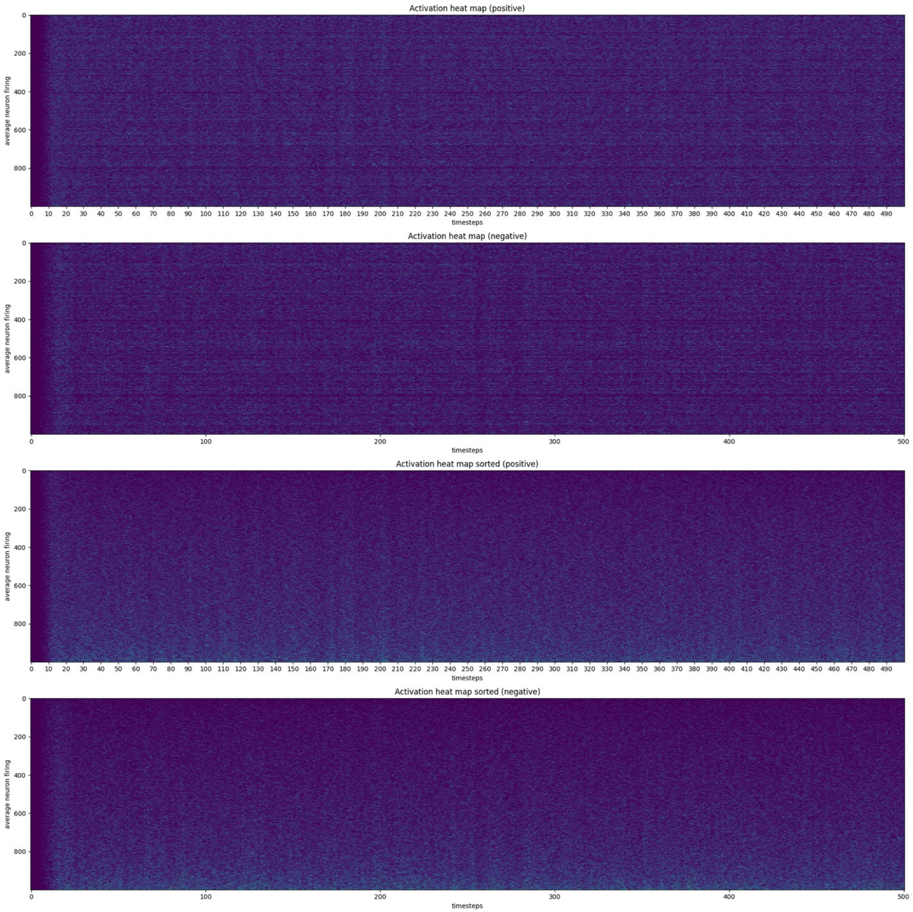
Methods

- Exploratory analsis of dataset
- Implemented a range of neural codings
- Trained a Logistic Regression for each coding
- Summarized regression performance

Exploratory analsis of dataset

Heatmap of neuron activations

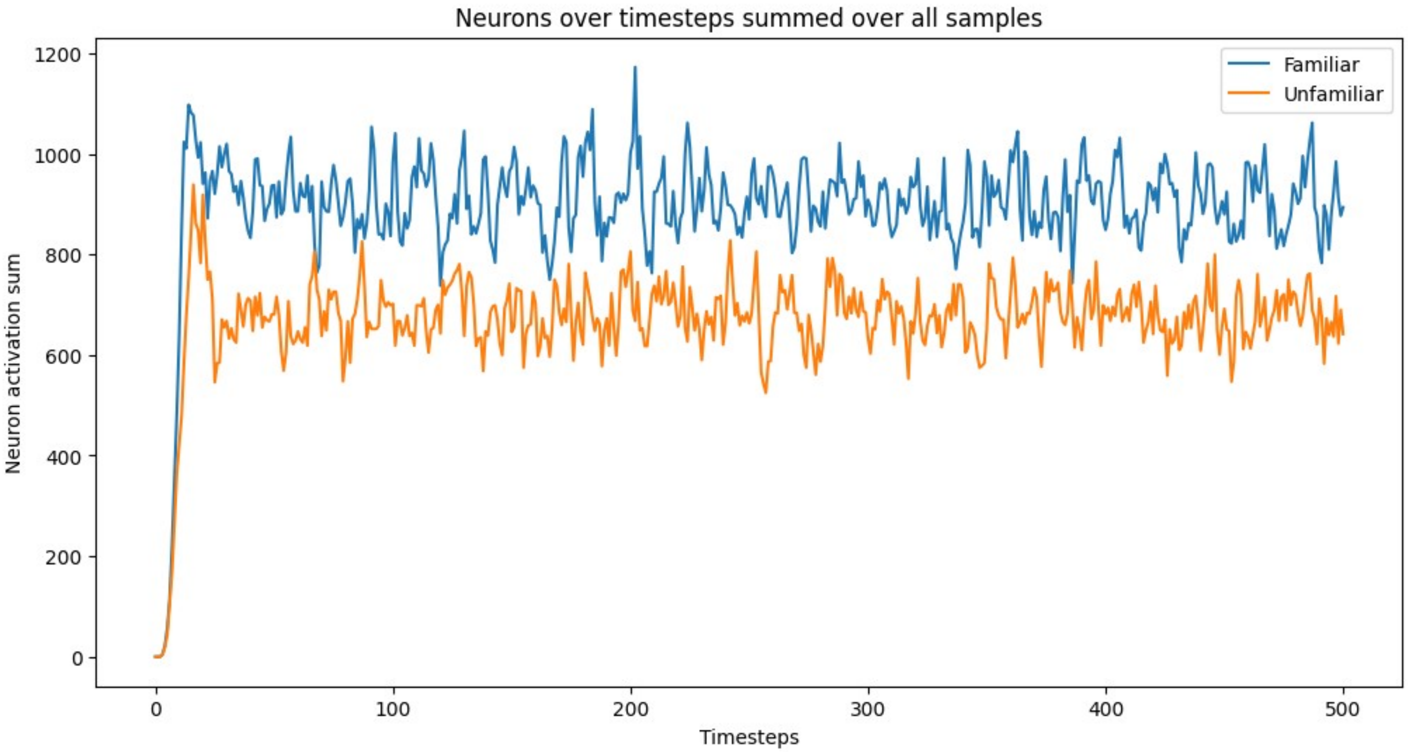
- Very noisy
- Begins with no acitivity
- Ramps up after short delay
- Slight vertical lines are visible
- Maybe there is correlation between neurons



Summed activity over all samples

Differences between conditions

- Familiar has more activity in total
- Maybe familiar has more variance



Different coding methods

Spike count rate

- Sum of spikes over all timesteps
- Represents amount of activity after stimulus

Time to first spike

- How many timesteps does it take until the first spike gets fired
- The activity of neurons starts slightly late for unfamiliar condition

Phase of firing

- Take all phases of each neuron after fft
- Might encode offsets of fire patterns

Frequency of highest amplitude

- Takes the frequency with the highest amplitude per neuron
- Could detect differences between the prominent frequency of conditions

- Population variance
- Take the variance of activations over all neurons
- Summed activity slide showed slight differences in variance

- Population average
- Average all neurons for every time step
- In summed activity slide there was a clear sum difference between conditions

Results

Okay performance:

- Spike count rate
- Time to first spike
- Population average

70% - 80% accuracy

Bad performance:

- Phase of firing
- Frequency of highest amplitude
- Population variance

40 - 60% accuracy